Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Musaab Ismail	musaab@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I want to build a website that helps me review the most used verbs (100 verbs), nouns (100 nouns), and expressions (20 expressions) in each of the languages that I know (Arabic, English, Spanish).

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

From my previous look at this, there are apps that help with reviewing your languages. These websites/apps offer multiple languages and mine differs from them by only having three languages, being extra simple, and testing your knowledge on things that you will actually hear people use in your daily life.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I think the above gives a sufficient explanation. But… I want this web application to be used as a review tool for languages (in my case, English, Arabic, and Spanish).

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

This program will assist in retaining user's knowledge about a specific language through textual input.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Answers to the questions being asked by the program (These questions vary depending on the mode of study that the user chooses). There are (most likely) three modes of study… translating between languages (nouns, verbs, and expressions), active recall flash cards (how many can you remember from a list of ten verbs/nouns), and identifying verbs and nouns in sentences. There will also be a list of words that acts as a dictionary.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Compare the input against a database and give feedback to the user.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

A checkmark representing correctness. A list of words when clicking the button linked to the dictionary of a specific language. And a number representing how many words you correctly remembered from the "active recall" list. (this last one might be replaced by the "chat with the machine" feature explained later in this document.)

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Nouns, verbs, and translations between two languages.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

The above are basically the same thing (in terms of code mechanism) and have the same level of importance. *The Details I need→which they all share in common: is it written correctly, is it a synonym, is it punctuated/written with white space/empty/etc.*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The fact that I might need to use gemini's API to allow for real time conversation between my website (AI under the hood) and the visitor. Also, visitors might input things that cause the website to malfunction so the testing-and-breaking of this website might be somewhat demanding/difficult.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will be using an LLM and I will copy and paste the code but not run it without first taking a

good look at it and that the components that I tell it to include are actually present in the code then I will run the code, if the code has an error then I will use my error tracing skills and debug my own code (I will also use AI for that probably.)

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Learn drafter deeply and use the website I built for practicing the languages that I know (and are forgetting).

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I told my friend and sister this idea and they expressed some interest in it (kind of). I did not ask for real feedback, I just asked if this is a viable idea for a website and should I continue with it (indirectly asked that).

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

I did not use AI for this. My friend and my sister did not suggest anything. They just expressed amazement (somewhat)

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

No pushback occurred.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

It didn't change, one of the people I was sitting close to said that I can use gemini's API because it gives about 1k requests per day (I think). The fact that I can integrate this into my code means that I can add a button that acts as a robust conversational feature…but this might not be the case if it is too complicated otherwise I might do that. This also brings the challenge that a user might overuse this tool and deplete all of the requests available from the API thus breaking the website or putting it in "time-out" mode.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I am using AI so I probably won't get stuck.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will look at the debugger panel to see where the problem is. Then, I will use my previous knowledge about Drafter to solve the problem. If I get stuck in the second step, then I will ask AI to help me solve the problem.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM to assist me with this assignment.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.